MILITARY SPECIFICATION

WRENCH: STRUCTURAL; BRIDGE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a wrench for bridge erection equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command. ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-171	- Finishing of Metal and Wood Surfaces.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM A 322 - Hot-Rolled Alloy Steel Bars.
ANSI/ASTM E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The wrench shall be as shown in figure 1 and as specified herein.

3.2 Material. Material shall be as specified herein (see 6.3 and 6.4).

3.2.1 Steel. Steel shall conform to ANSI/ASTM A 322, UNS G13400, condition normalized and tempered (NT), minimum yield strength of 100,000 pounds per square inch (psi).

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3.3 Design. The wrench shall be designed to afford a comfortable handgrip. The nut-and-bolt head engaging surfaces of the wrench shall be finished in a smooth and well-defined manner. The forward end shall be chamfered or rounded to provide a lead.

3.4 Head size. The wrench shall be 2.0625 inch. The wrench opening shall be not less than 2.074 inches and not more than 2.088 inches.

3.5 Parallelism. Tolerance for the parallelism of the nut-and-bolt head engaging surface shall be 0.005 inch.

3.6 Hardness. The wrench head shall be hardened to a Rockwell hardness of not less than C35 nor more than C55.

3.7 Finish. Finish shall be one of the following as specified (see 6.2):

a. Zinc plate conforming to MIL-STD-171, finish No. 1.9.2.2;

b. Hot phosphate conforming to MIL-STD-171, finish No. 5.3.1.2;

c. Oxide conforming to MIL-STD-171, finish No. 3.3.1.1.

3.8 Marking. The wrench shall be marked on one of the faces, in a plain and permanent manner with the size of jaw opening, and in accordance with MIL-STD-130.

3.9 Workmanship. Wrenches shall be free from rust, fins, burrs, external sharp or rough edges or surfaces, and defects which may impair serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perfor any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirement

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured and examined in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspection requirements specified herein ar classified as follows:

a. Quality conformance inspection (see 4.3).

b. Inspection of packaging (see 4.5).

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4.3 Quality comformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. The AQL shall be 0.4 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.3 Tests.

4.3.3.1 Samples. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.5.1. AQL shall be 2.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The wrenches shall be examined as specified herein for the following defects:

Major

101. Wrench not as shown in figure 1.
102. Material not as specified.
103. Design not as specified.
104. Head size not as specified.
105. Parallelism not as specified.
106. Finish not as specified.
107. Workmanship not as specified.

Minor

201. Marking missing, incomplete, illegible, or not as specified.

4.4.2 Test.

4.4.2.1 Hardness. The wrench shall be tested in accordance with ANSI/ASTM E 18. Nonconformance to 3.6 shall constitute failure of the test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack. The preservation, packing and marking of the wrenches for level A or B shall be examined to determine compliance with the quality assurance provisions of PPP-P-40. Commercial preservation, packing and marking shall be examined to determine compliance with MIL-STD-1188.

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5. PACKAGING

5.1 Preservation, packing and marking. The preservation, packing and marking for level A or B shall be in accordance with PPP-P-40. The preservation, packing and marking for commercial shall be in accordance with MIL-STD-1188. The degree of preservation and packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The wrenches are intended for use in the erection equipment for single lane, fixed, panel bridges.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of finish required (see 3.7).

c. Degree of preservation and degree of packing required (see 5.1).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

6.4 Materiel deterioration prevention and control (MADPAC). The contracting officer should consider for inclusion in all applicable invitations for bid, requests for proposal, and contracts the following statement:

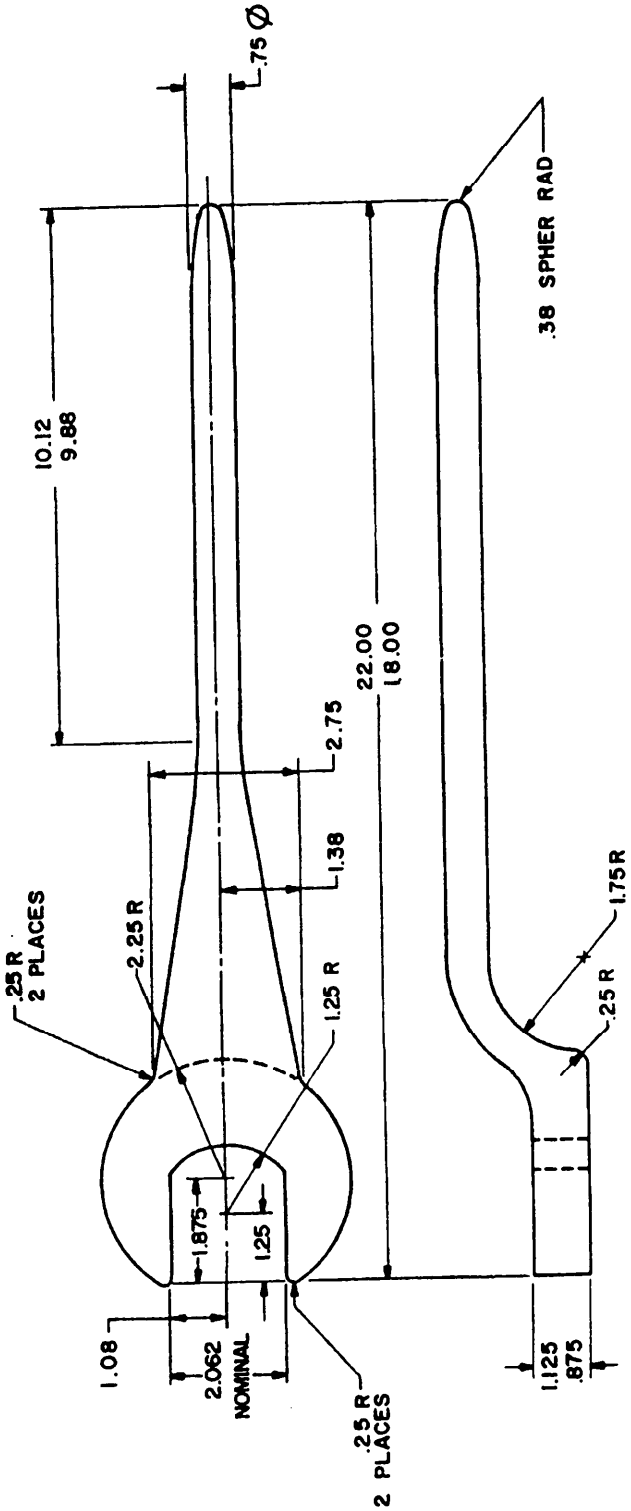
"The technical proposal shall provide assurance that all system/item parts shall be fabricated from compatible materials, suitably finished, to provide protection from corrosion and other forms of materiel deterioration. The assurance shall be in the form of a finish plan, utilizing state-of-art technology, which clearly identifies the specific finish or treatment to be used on the various components and assemblies, to provide protection against the various forms of materiel deterioration in the applicable operating and storage environments to which they may be exposed." (see 3.2).

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NOTE:
1. ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED
2. TOLERANCES, UNLESS OTHERWISE SHOWN: XX ± .06 XXX ± .030

FIGURE 1. Wrench: structural; bridge.

X-2789A

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-W-52799A(ME) Wrench: Structural; Bridge

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